Research Findings

1. HTTP State Management and User Authentication

HTTP is a stateless protocol, meaning each request from a client to a server is treated independently without inherent knowledge of previous interactions. To manage state across multiple request-response cycles, applications often utilize mechanisms such as cookies, sessions, and tokens. Cookies store user data on the client side and are sent with each request, allowing servers to recognize users. Sessions, on the other hand, store user data on the server side, often identified by a unique session ID held in a cookie. This approach is commonly used for user authentication, where upon login, a session is created to track the user's state. Additionally, modern applications may implement token-based authentication, such as JWT (JSON Web Tokens), which allows for secure, stateless user verification while maintaining session information across requests.

2. Django Database Migrations to MariaDB

When deploying a Django application using a server-based relational database like MariaDB, the migration process involves several key steps. First, you need to define your models in Django's `models.py` file. After defining or modifying the models, the command `python manage.py makemigrations` generates migration files that describe the changes in the database schema. To apply these migrations to the MariaDB database, you execute `python manage.py migrate`, which updates the database according to the migration files. It's essential to ensure that the database connection settings in `settings.py` are correctly configured for MariaDB. This includes specifying the database name, user, password, and host. Additionally, running migrations can be monitored through the Django admin interface or by checking the database directly to ensure that all changes are applied accurately.